Project 1: Sheep

MATH3060 – Statistical Data Analysis & Probability Stream

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Abstract

This report investigates optimal decision-making strategies for the *Sheep* dice game, a two-player race to 100 points with compound risk mechanisms. We develop and compare four strategies of increasing sophistication: a naive fixed-threshold policy, an expectation-based model, an optimal dynamic programming (DP) solution, and a simplified DP-Median approximation. Monte Carlo simulations reveal that the DP-optimal strategy achieves a 70.8% win rate against the naive baseline, with a first-mover advantage of approximately 51.4%. All observed differences are statistically significant, apart from DP-Median vs DP-Optimal, where the confidence intervals overlap. As a result, and as a practical recommendation, we propose the DP-Median strategy, which condenses the optimal policy into a compact lookup table, achieving near-optimal performance while remaining feasible for human play.

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1 Introduction

The *Sheep* dice game is a two-player race to 100 points in which players repeatedly decide whether to roll two dice or secure the points accumulated during the current turn. It belongs to the broader family of *Pig* games—classical dice games where players build points but lose them if an unfavourable roll occurs. In the standard *Pig* game, rolling a 1 ends the turn with zero gain. However, *Sheep* introduces a substantially more volatile risk structure. Rolling a single six immediately ends the turn and forfeits the turn total, while rolling double six resets the player's entire cumulative score.

Neller and Presser [1] demonstrated that the standard Pig game admits an exact solution via dynamic programming, computing optimal hold thresholds for all possible game states. Gámez et al. [2] extended this analysis to variants with modified dice rules, showing how small rule changes significantly alter optimal play. More recently, Zhu and Chen [3] applied Monte Carlo methods to quantify risk–reward trade-offs in stochastic dice games. We build on these approaches to analyse the *Sheep* variant, which introduces the additional catastrophic risk of losing all points on double sixes.

This project addresses the central strategic question of the *Sheep* game: deciding when a player should continue rolling and when they should hold. To investigate this, we examine four decision rules of increasing complexity. We begin with a simple fixed-threshold strategy and a probabilistic expected-value rule, then compute the fully optimal dynamic-programming policy and extract a simplified "DP-Median" threshold that captures its dominant structural features. These strategies are compared through simulation to evaluate their risk profiles, efficiency, and practical usability. Since the optimal policy is impractical for players to follow, the report ultimately proposes an intuitive and actionable strategy that retains the essential logic of optimal play.

The remainder of this report is structured as follows. Section 2 covers the methodology for each strategy. The strategies are then compared in section 3 and section 4 explores extensions to the *sheep* game. Section 5 offers some concluding remarks.

2 Methods

2.1 Naive baseline

Before introducing more sophisticated optimisation techniques, we establish a simple baseline by examining how a player performs under a fixed, non-adaptive rule. We first outline the fundamental assumptions of the game, followed by an additional assumption specific to the naive strategy.

1. **General modelling assumptions:** The game is played with two fair six sided dice, therefore the probability of landing on each face is uniformly distributed with $P = 1/6$. Players seek to maximise their total score, up to 100.

2. **Additional assumption for the naive strategy:** The player's decision is based solely on the current turn total, with no consideration of cumulative scores or previous game states.

Under these assumptions, we considered two simple types of behaviour, first is **Single-roll policy** ($K = 0$)the player rolls only once per turn and then holds immediately. Secondly a **Fixed-threshold policy (Hold-at-K)**, where the player keeps rolling until the current turn total reaches or exceeds K. This simplified example helps us understand the basic dynamics of the game before moving on to more advanced models.

To investigate how the choice of a fixed threshold influences player performance, we conducted Monte Carlo simulations of the Sheep dice game. We first tested a wide range of thresholds from 0 to 30 to observe the overall performance pattern. In each case, two identical players followed the same Hold-at-K rule, and 20,000 independent games were simulated to estimate the probability of the first player winning. After this initial exploration, a smaller group of thresholds was selected for more detailed testing. Each of these values was repeated 100 times with different random seeds but under the same game settings, to check whether the estimated win rates were stable across runs and to compare the performance of different thresholds.

To validate the reliability of our Monte Carlo estimates, we tracked win rate convergence as the number of simulated games increased. Figure 1 demonstrates that estimates stabilise rapidly, with 95% confidence intervals narrowing substantially by 5,000 games and reaching acceptable precision at 10,000 games. This convergence behaviour justifies our choice of 10,000 simulations per matchup for the main analysis.

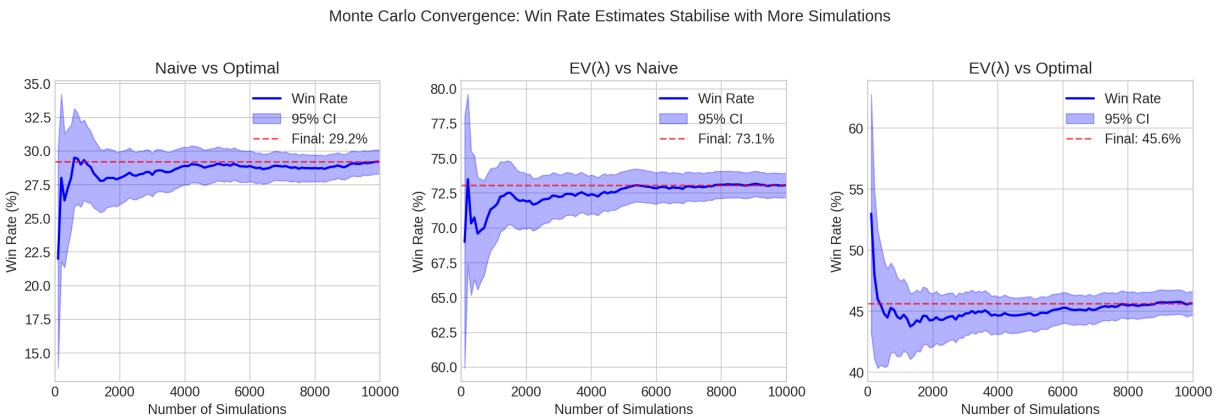

Figure 1: Monte Carlo convergence: win rate estimates stabilise as the number of simulations increases. Shaded regions show 95% confidence intervals computed using the standard formula.

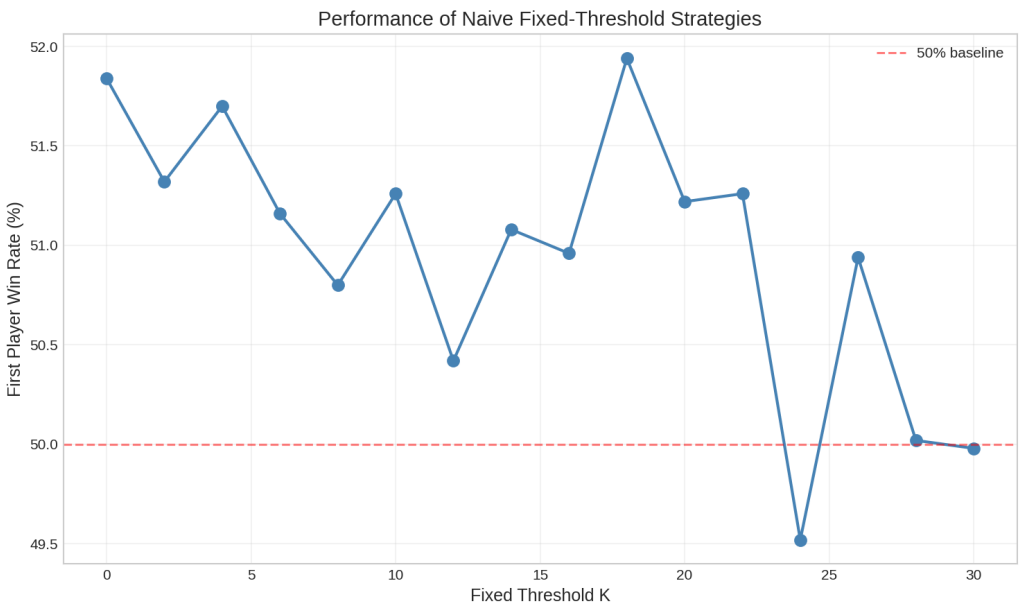

Figure 2: First-player winning probability for different fixed-threshold values K. The red dashed line indicates the 50% baseline.

The initial simulations across a wide range of thresholds ($K = 0 - 30$) show that the estimated winning probabilities for each K fluctuated slightly between runs due to randomness in dice rolls.

Figure 2 shows the first-player winning probability when both players follow the naïve fixed-threshold rule. A mild downward trend appears as K increases, but overall the win rate remains in a narrow band (50%–52%). In particular, very small thresholds such as K=0 and K=5 tend to perform marginally better in this run, and occasional peaks also appear around K=15–20. However, these fluctuations are inconsistent between simulations and well within Monte-Carlo variability, meaning no specific fixed threshold consistently outperforms the others. Given the absence of a clearly superior fixed threshold, we choose K=0 as the baseline strategy for later sections. It corresponds to the simplest interpretable rule ("roll once then hold") and provides

a natural comparison point for more advanced strategies.

2.2 Expected Value Based Strategy

The naive threshold strategy does not consider the total score and risk of rolling. To account for these factors, we introduce an expectation-based model, which remains simple and intuitive. This examines the expected change in total value by rolling or holding $E[\Delta(S+T)]$, where S is the players total score and T is the players turn total. The player's state can be denoted by (S, T).

In addition to the general assumptions in Section 2.1, the expectation model makes one further simplification: decisions are evaluated locally and depend only on the current turn total.

For any roll, three outcomes may occur:

1. **Double six ($P = 1/36$)**: the player loses all accumulated points and returns to (0,0). Out of all $6^2 = 36$ combinations of dice rolls there is only one (6,6) for this to occur. Given that we roll this combination $\Delta(S+T) = -(S+T)$.

2. **Single six ($P = 10/36$)**: the player loses the turn total and moves to $(S, 0)$. There are 10 combinations of rolls where this occurs. If we roll this combination $\Delta(S+T) = -T$.

3. **Safe roll ($P = 25/36$)**: the player adds the sum of the dice to their turn total. There are 25 combinations of dice rolls resulting in an addition to the turn total. $E[\Delta(S+T)|Safe\ Roll] = 2 \times \frac{5-1}{2} = 6$ equivalent to randomly sampling from to independent distributions $U(1, 5)$.

Comparing rolling and holding:

$$\begin{aligned} E[\Delta(S+T) \mid Roll] &= \frac{1}{36}[-(S+T)] + \frac{10}{36}[-T] + \frac{25}{36}[6] \\ &= \frac{150 - S - 11T}{36}. \end{aligned} \tag{1}$$

$$E[\Delta(S+T) \mid Hold] = 0$$

Under a greedy process we maximise the immediate expected reward without considering the future process. Therefore, we can choose the decision roll vs hold which maximises our expected value for this turn. This can serve as an intuitive and playable strategy.

We roll when:

$$E[\Delta(S+T) \mid Roll] > E[\Delta(S+T) \mid Hold] = 0$$

OR

$$\frac{150 - S - 11T}{36} > 0$$

Otherwise we hold.

2.3 Extended Model: EV(λ) with Opponent Awareness

The issue with the expected value based strategy is that it does not account for the opponents current score. We can introduce a score difference term, aiming to change the approach to play when we are far behind or ahead the opponent. Intuitively, we want to play risk on when trailing, but safe when ahead.

$$\text{Decision value} = \frac{150 - S - 11T}{36} + \lambda(S - R) \tag{2}$$

Where S is the player's permanent score, T is the current turn total, R is the opponent's score and λ is a tuning parameter. We roll if decision value > 0, otherwise we hold.

Interpretation of λ.

- $\lambda < 0$: Play **safer** when ahead ($S > R$), **riskier** when behind
- $\lambda > 0$: Play **riskier** when ahead, **safer** when behind
- $\lambda = 0$: Ignore opponent's score entirely (reduces to basic EV model)

Using a grid search over the range $-0.2 < \lambda < 0.2$, we simulate $100,000$ rounds and compare the first player win probability. We find that the optimal parameter was determined to be $\lambda = -0.02$ as seen in figure 3. This negative value reflects sound game-theoretic intuition: when leading, preserve the advantage by holding earlier; when trailing, take calculated risks to catch up.

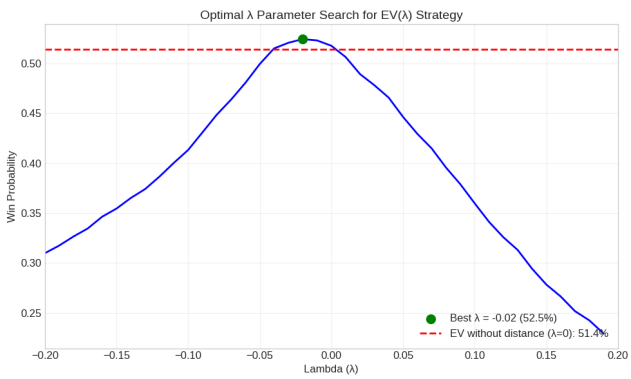

Figure 3: Optimal Lambda

The EV(λ) strategy thus provides a simple and interpretable rule, which adapts to a competitive context, while remaining computationally trivial compared to a dynamic programming approach. Figure 4 shows the EV(λ) decision surface with $\lambda = -0.02$. The horizontal axis

is the player's cumulative score s, and the vertical axis is the turn total t. The green region indicates states where rolling is favourable, while the red region indicates holding.

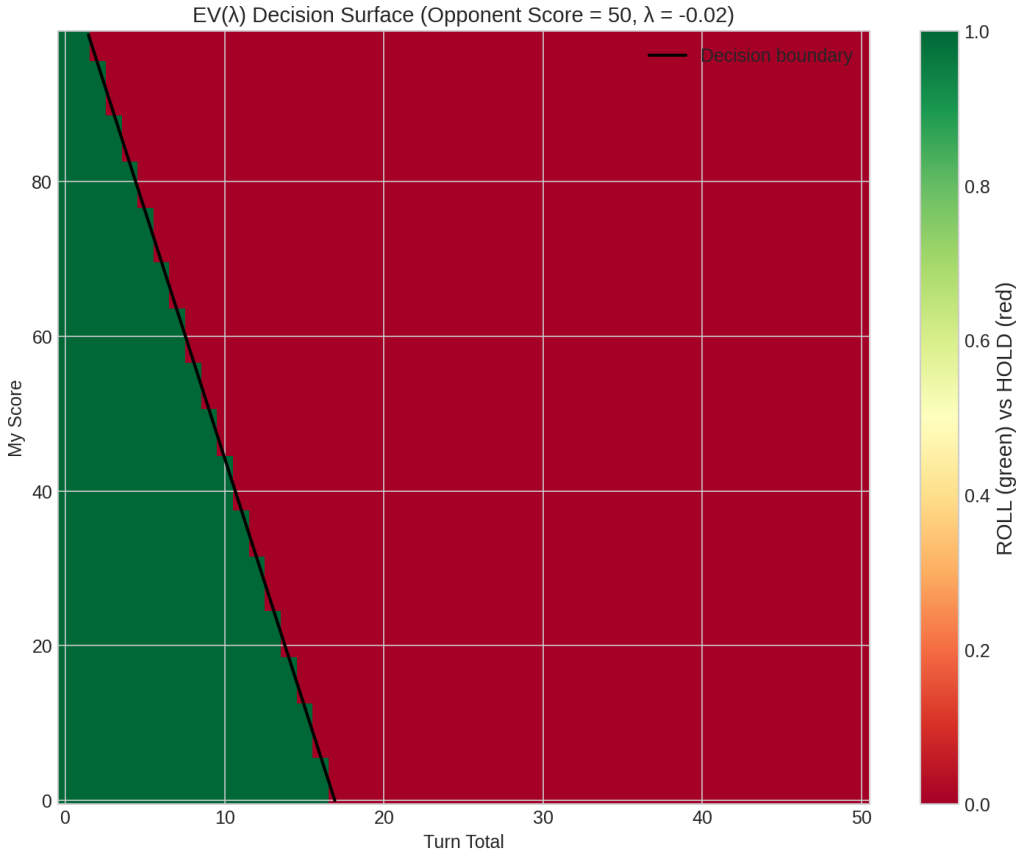

Figure 4: Expected Value Boundary for Roll/Hold Decision. The decision surface shows when to ROLL (green) versus HOLD (red) based on current score and turn total.

The decision threshold declines linearly with S, indicating that players should roll longer when behind and hold earlier as they approach 100 points. This reflects the intuitive trade-off between risk and reward captured by the expected value model. However, although the model provides a clear analytical threshold, it simplifies the game by by taking a greedy approach, ignoring future implications. This limitation motivate the more comprehensive dynamic programming analysis developed in the next section.

2.4 Dynamic Programming Approach

In Sheep, each decision to hold or roll depends only on the current scores and turn total—the game has no memory of how players reached that state. This property lets us compute the exact probability of winning from any position by working backwards from winning states. We use value iteration: starting with known outcomes (reaching 100 points wins), we repeatedly update win probabilities until they converge.

We assume both players play optimally and know each other's strategy. Under these conditions, we can compute a single value function $V(S, R, T)$ giving the win probability from each state,

and derive the optimal policy directly from it. The game state is defined as (S, R, T), where S is the player's cumulative score, R is the opponent's score, and T is the current turn total. At each state, the player selects an action $A \in \{\text{Hold}, \text{Roll}\}$, and the next state follows from the transition rules below. This triplet fully characterises the game and determines all subsequent transitions.

Hold. If the player holds, the turn total T is added to their score. If $S + T \geq 100$, the player wins immediately. Otherwise, control passes to the opponent, resulting in the new state $(R,\ S + T,\ 0)$.

Roll. If the player rolls, one of three probabilistic outcomes occurs:

1. **Double six** (1/36): both dice show six; the player loses all accumulated points, and the opponent takes control in state $(R,\ 0,\ 0)$.
2. **Single six** (10/36): one die shows six; the turn total is lost but S is retained, giving state $(R,\ S,\ 0)$.
3. **Safe roll** (25/36): neither die shows six; the player gains the sum of the dice, yielding the new state $(S,\ R,\ T + d_1 + d_2)$.

Let X denote the sum of the two dice, taking values $\{2, 3, 4, 5, 6, 7, 8, 9, 10\}$. After a safe roll, the player transitions to the updated state $(S, R, T + X)$.

The state space consists of all triples (S, R, T) where both scores are below 100 and $T \geq 0$. In practice, we cap T at 50 since higher turn totals are extremely rare. This gives roughly $100 \times 100 \times 51 \approx 510{,}000$ states to evaluate. At each state, the player chooses either "Hold" or "Roll".

Let $V(S, R, T)$ denote the probability that the current player eventually wins from state (S, R, T) under optimal play. The Bellman equation is

$$V(S, R, T) = \begin{cases} 1, & \text{if } S + T \geq 100, \\ 0, & \text{if } R \geq 100, \\ \max\{H(S, R, T),\ R(S, R, T)\}, & \text{otherwise.} \end{cases} \tag{3}$$

Hold. If the player chooses to hold, the turn total T is banked and roles swap:

$$H(S, R, T) = 1 - V(R,\ S + T,\ 0). \tag{4}$$

Roll. If the player rolls, the outcome falls into one of three probabilistic cases:

$$R(S,R,T) = \begin{cases} \text{Single six } (p_s = 10/36): & 1 - V(R,S,0), \\ \text{Double six } (p_d = 1/36): & 1 - V(R,0,0), \\ \text{Safe roll } (p_r = 25/36): & \sum_{s=2}^{10} \frac{w_s}{25} V(S,R,T+s), \end{cases}$$

where the weights (w_s) give the number of two-dice outcomes summing to s:

$$(w_2, w_3, w_4, w_5, w_6, w_7, w_8, w_9, w_{10}) = (1,2,3,4,5,4,3,2,1).$$

Combined form. Collecting all components, the roll value can be expressed compactly as

$$R(S,R,T) = p_s\big[1 - V(R,S,0)\big] + p_d\big[1 - V(R,0,0)\big] + \sum_{s=2}^{10} \frac{w_s}{36} V(S,R,T+s). \tag{5}$$

This completes the Bellman optimality relation for the optimal winning probability $V(S,R,T)$. Analytical solutions to the Bellman equations are intractable, so numerical value iteration is used to approximate $V(S,R,T)$. We initialise the value function with:

$$V_0(S,R,T) = 0.5,$$

and update it iteratively via

$$V_{k+1}(S,R,T) = \max\{V_{\text{hold}}(S,R,T),\, V_{\text{roll}}(S,R,T)\}.$$

We stop iterating when the maximum change across all states falls below $\varepsilon = 10^{-8}$:

$$\max_{S,R,T} |V_{k+1}(S,R,T) - V_k(S,R,T)| < \varepsilon.$$

In our implementation, this convergence criterion is met after 61 iterations. Convergence was monitored through the residual $\Delta_r = \|V_{r+1} - V_r\|_\infty$, which decayed geometrically, confirming numerical stability. After convergence, the optimal hold threshold is defined as

$$T^*(S,R) = \min\{\, T : V_{\text{hold}}(S,R,T) \geq V_{\text{roll}}(S,R,T)\,\},$$

representing the smallest turn total at which holding becomes preferable to rolling.

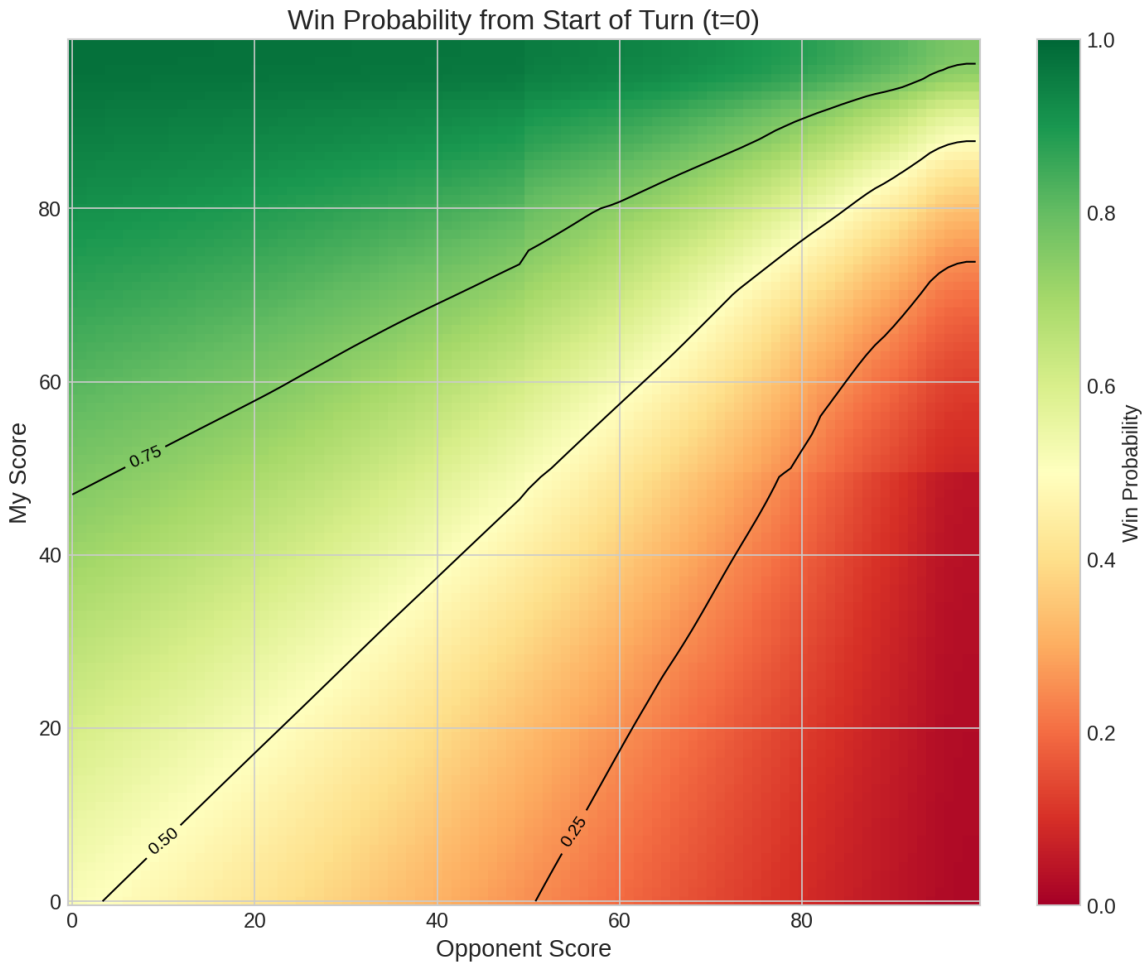

Figure 5: Win Probability Surface from Start of Turn ($t = 0$). Contour lines show 25%, 50%, and 75% win probability.

The value iteration converges within 61 iterations with residual $|\Delta V|$ decreasing almost exponentially from 10^{-1} to below 10^{-6}, demonstrating fast and stable convergence of the scheme. Figure 5 shows the resulting win probability surface at $t = 0$. The final value $V(0, 0, 0) \approx 0.514$ indicates a slight inherent advantage for the first player under optimal play.

Figure 6 visualises the optimal hold-at-T threshold $T^*(S, R)$ obtained from the dynamic programming model. Each grid cell corresponds to a game state (S, R), with the player's score S on the vertical axis and the opponent's score R on the horizontal axis. The colour intensity indicates the optimal turn total T^* at which holding becomes preferable to rolling, with brighter regions representing higher thresholds (more aggressive play).

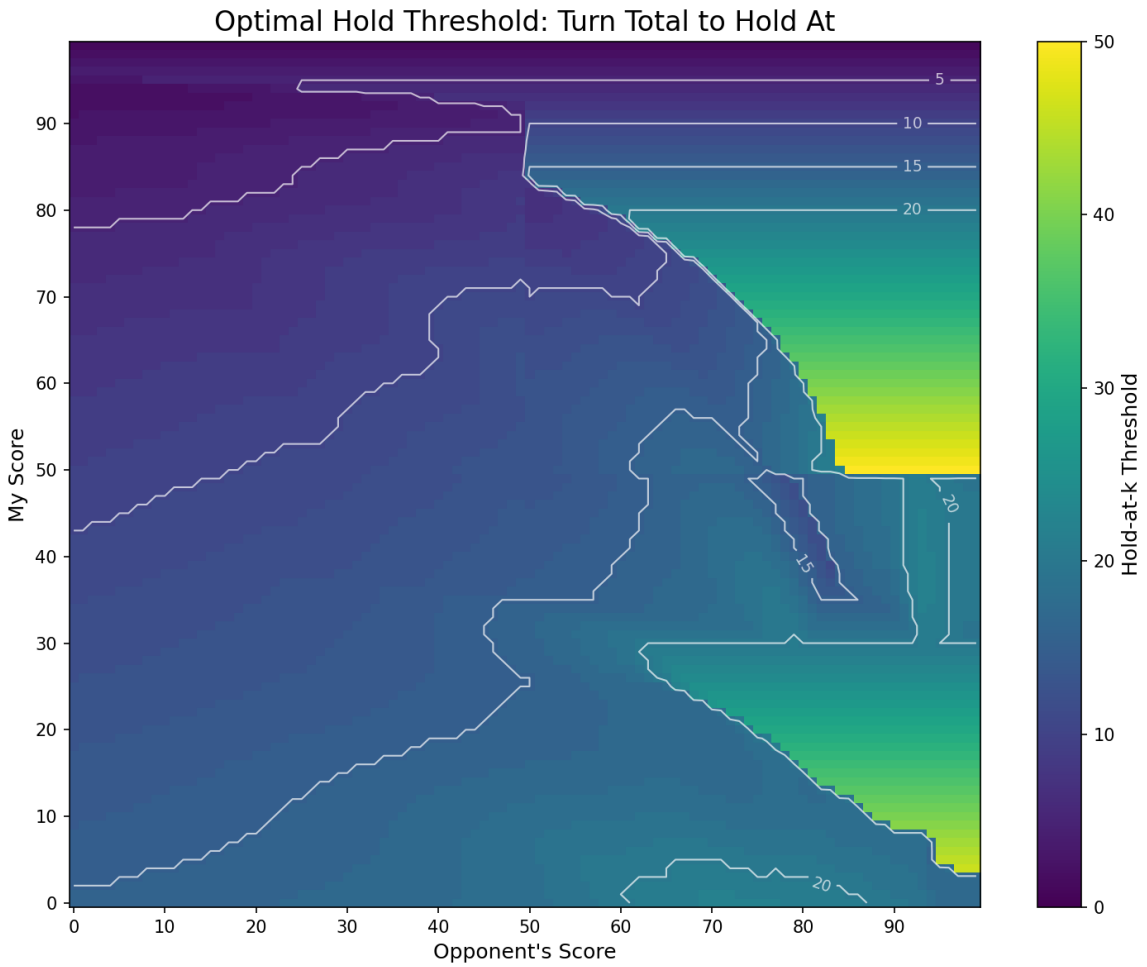

Figure 6: Optimal hold threshold $T^*(S, R)$ for each game state. Brighter colours indicate higher thresholds (roll longer before holding). Contour lines mark key threshold values.

The heatmap shows that $T^*(S, R)$ generally decreases as the player's score increases. Near the winning region ($S + T \geq 100$), the optimal policy becomes increasingly conservative, reflecting the high penalty of single-six or double-six outcomes. In contrast, players who are far behind adopt higher thresholds to maximise comeback potential. A smooth transition appears near the diagonal $S \approx R$, indicating balanced competitive states where risk and reward are more evenly traded off.

Figure 7 shows the resulting HOLD/ROLL decisions at specific turn totals, illustrating how the decision boundary shifts as the turn total increases.

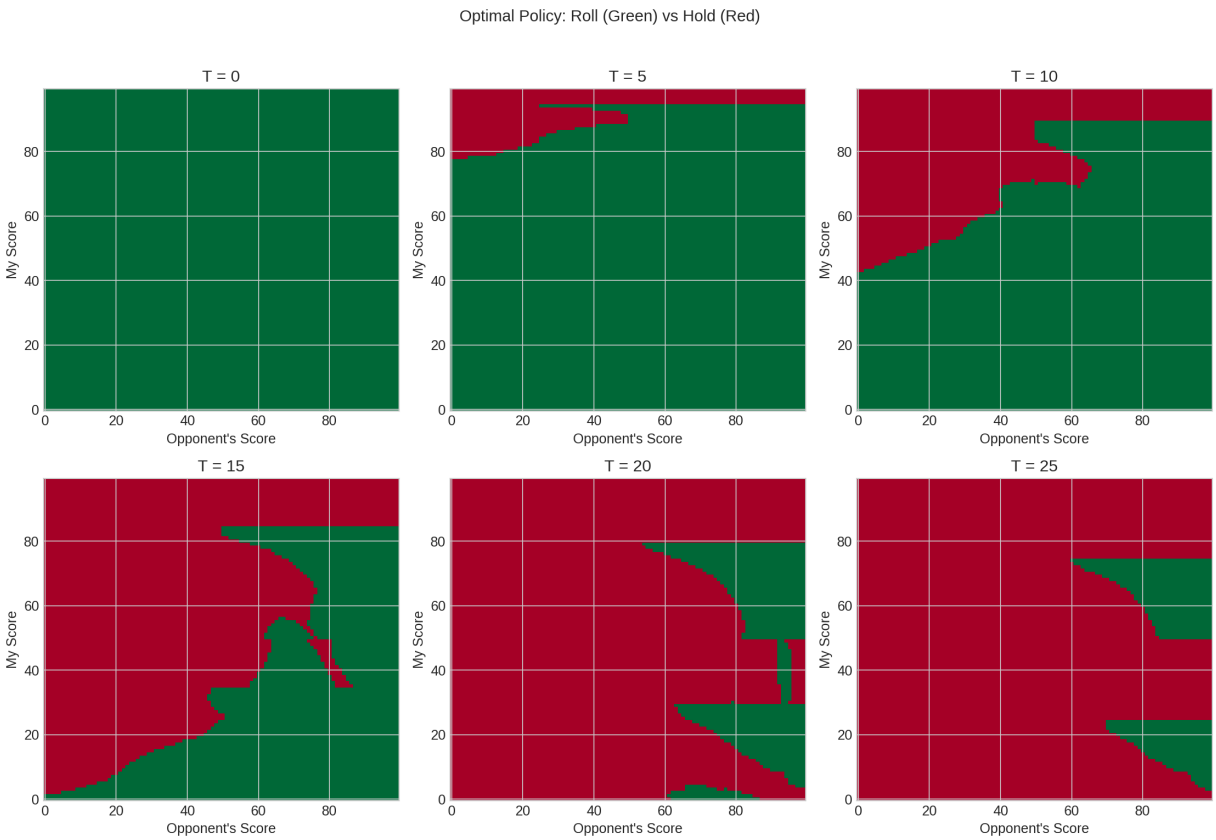

Figure 7: Optimal policy at different turn totals: HOLD (green) vs ROLL (red). As turn total increases, the HOLD region expands, reflecting the growing risk of continuing to roll.

The staircase structure along the contour lines arises from the discrete dice sums in the model. Because the Sheep Dice Game includes two failure modes—single six (loss of T) and double six (loss of all points)—the effective risk is higher than in the single-die Pig game, leading to lower optimal thresholds overall. The heatmaps therefore highlight how optimal play adapts aggressiveness and caution according to the score difference.

Dynamic programming provides the exact optimal strategy for the Sheep Dice Game by evaluating all possible states and applying the Bellman equations, offering a complete and theoretically optimal solution. Figure 8 compares the decision boundaries of the EV(λ) strategy with the optimal DP policy at different opponent scores, illustrating how both strategies adapt their aggressiveness based on game state.

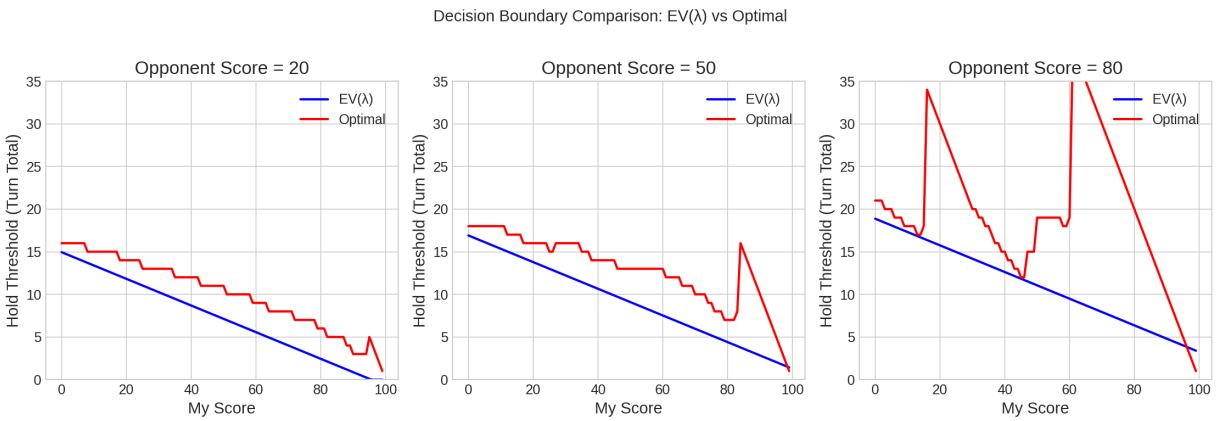

Figure 8: Decision boundary comparison: EV(λ) vs Optimal policy at opponent scores 20, 50, and 80. The threshold indicates the turn total at which each strategy switches from ROLL to HOLD.

However, the resulting policy is highly state-dependent and too complex for practical play, making the optimal strategy difficult for humans to implement without computational assistance.

2.5 Constructing the DP-Median Playable Strategy

Because the full DP strategy depends on a large and detailed threshold table, it is not practical for players to use during an actual game. For this reason, we look for a simpler way to approximate the DP results. We first divide the player's current total score into four stages, as shown in Table 1.

Score range (S)	**Game phase**	**Strategic tendency**
0–39	Early game	Low risk; build base score, roll moderately.
40–69	Mid game	Balanced risk–reward; thresholds near fixed optima.
70–89	Late game	More cautious; reduce threshold to avoid double–6 loss.
90–99	End game	Very conservative; cap threshold at remaining points to win.

Table 1: Score phases and corresponding strategic tendencies.

Here we define the score difference as $d = S - R$, where S is the player's current score and R is the opponent's score. In addition, the score difference is grouped into five broad ranges: $d \leq -20$, $-19 \leq d \leq -10$, $-9 \leq d \leq 9$, $10 \leq d \leq 19$, and $d \geq 20$. For each (S, d) category, we collect the corresponding DP-optimal thresholds $T^\star(S, R)$ and take the median value within that group. After rounding these medians to simple integers, we obtain a small lookup table that can be used directly during play. The resulting DP-Median strategy is summarised in Table 2.

My total score s	$d \leq -20$	$d \in [-19, -10]$	$d \in [-9, 9]$	$d \in [10, 19]$	$d \geq 20$
[0,39]	19	15	14	13	12
[40,69]	37	16	13	12	10
[70,89]	27	23	18	13	7
[90,99]	16	16	6	5	4

Table 2: Playable Threshold Table $H(s, d)$ constructed using DP-Median aggregation.

3 Results

3.1 Comparison of Four Strategies

We evaluate four strategies through Monte Carlo simulation of 10,000 games per matchup. To quantify the precision of our estimates, we compute 95% confidence intervals using the standard formula $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$. This interval is derived from the normal approximation to the binomial distribution, which is highly accurate when $np \geq 5$ and $n(1-p) \geq 5$. With $n = 10{,}000$ games and observed proportions ranging from 0.27 to 0.73, this condition is easily satisfied (smallest expected count = 2,700 $\gg$ 5). For our sample size, the margin of error is approximately $\pm 1\%$, providing precise estimates of the true win rates.

Matchup	**Strategy**	**Win Rate**	**95% CI**
Naive vs Optimal	Naive	29.2%	[28.3%, 30.1%]
	Optimal	70.8%	[69.9%, 71.7%]
EV(λ) vs Naive	EV(λ)	73.1%	[72.2%, 74.0%]
	Naive	26.9%	[26.0%, 27.8%]
EV(λ) vs Optimal	EV(λ)	45.6%	[44.7%, 46.6%]
	Optimal	54.4%	[53.4%, 55.3%]
DP-Median vs Naive	DP-Median	73.8%	[72.9%, 74.7%]
	Naive	26.2%	[25.3%, 27.1%]
DP-Median vs EV(λ)	DP-Median	57.0%	[56.1%, 58.0%]
	EV(λ)	43.0%	[42.0%, 43.9%]
DP-Median vs Optimal	DP-Median	49.1%	[48.2%, 50.1%]
	Optimal	50.9%	[49.9%, 51.8%]

Table 3: Win rates with 95% confidence intervals from 10,000 simulated games per matchup.

The DP strategy performs the best overall, achieving a 70.8% win rate against the Naive policy. The EV(λ) vs Optimal matchup shows closer competitive balance between these sophisticated strategies.

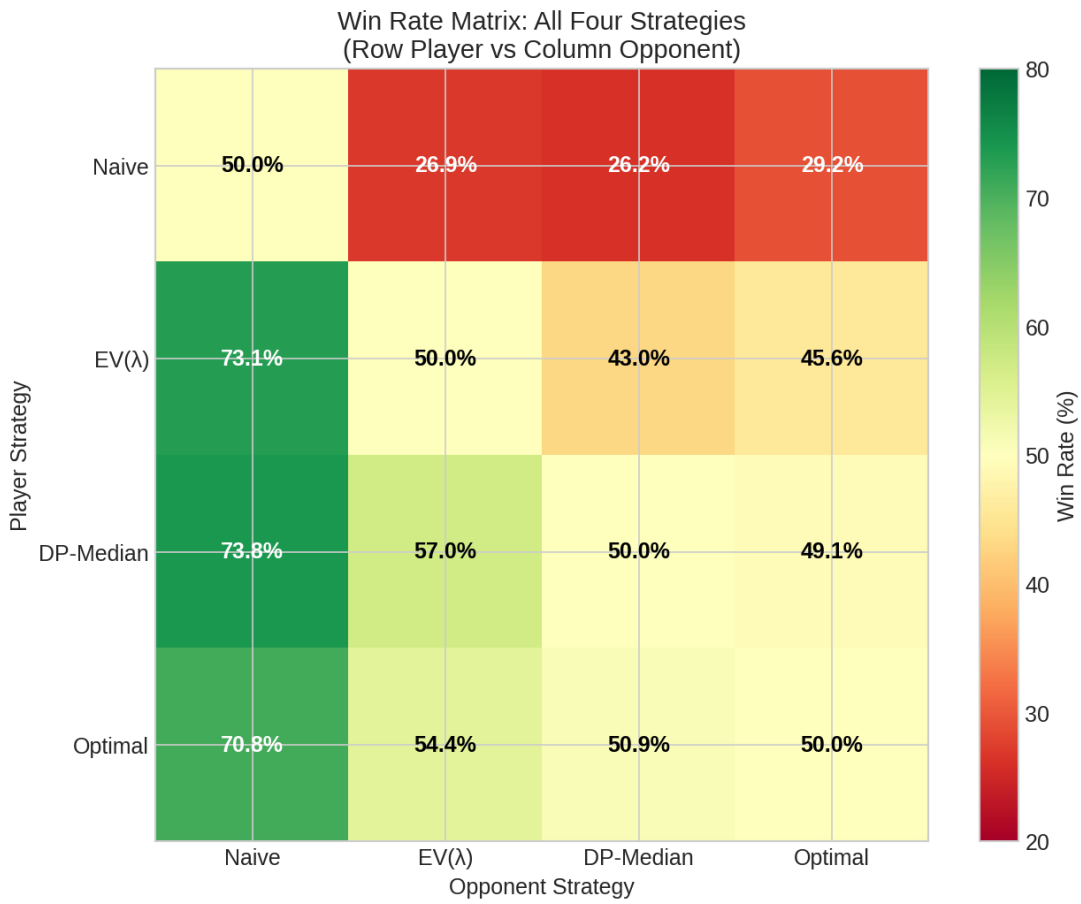

Figure 9: Win Rate Matrix for all four strategies. Each cell shows the row player's win percentage against the column opponent. DP-Median achieves near-optimal performance while remaining practically implementable.

Figure 9 presents the complete head-to-head results for all four strategies. The ranking is clear: Optimal > DP-Median > EV(λ) > Naive. Notably, DP-Median performs within 2–3 percentage points of the full DP-optimal policy across all match ups, validating its effectiveness as a practical approximation.

Interestingly, the EV(λ) model with $\lambda = -0.02$ achieves a 73.1% win rate against the naive baseline, slightly outperforming even the dynamic programming strategy. This reflects the difference between optimal and exploitative play. The dynamic programming strategy is optimal against optimal play but the naive strategy is constantly conservative. The EV(λ) model maintains moderate conservativeness with an average of 1.74 rolls per turn, taking advantage of the naive strategy's over conservativeness, whereas the DP approach makes more aggressive maneuvers (2.08 average rolls per turn), which are unnecessary against an extremely conservative opponent, causing more catastrophic failures (6's and double 6's). Due to the strategy's greedy approach, not accounting for future game states, the EV(λ) strategy still under performs when versing more complex, dynamic programming based solutions.

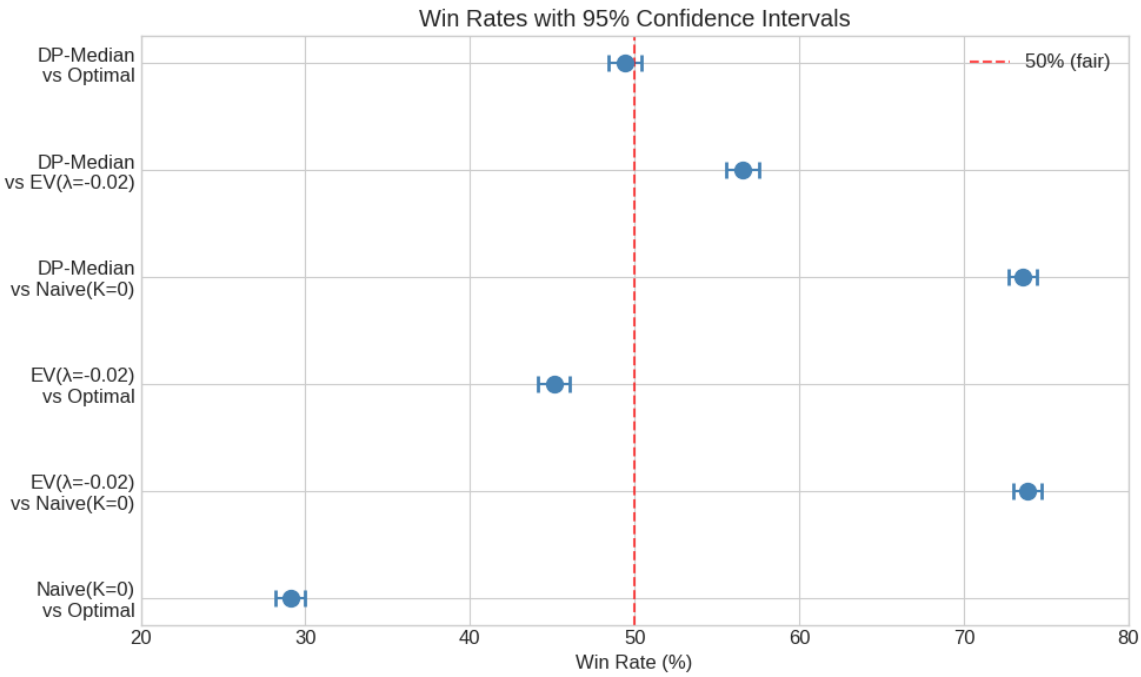

Figure 10: Win rates with 95% confidence intervals. The vertical dashed line indicates the 50% fair game threshold. Non-overlapping intervals indicate significant differences between strategies.

Among the more playable methods, both Expectation and DP-Median outperform Naive and remain relatively close to the DP benchmark. In their direct match-up, DP-Median shows a consistent advantage: it wins 57.0% of games against EV(λ), while EV(λ)'s win rate in the opposite direction is only 43.0%. The non-overlapping confidence intervals for EV(λ) vs Naive (73.1%, CI: [72.2%, 74.0%]) and Optimal vs Naive (70.8%, CI: [69.9%, 71.7%]) confirm that EV(λ) exploits the Naive baseline slightly better than Optimal does.

3.1.1 Game Length and Turn Outcome Analysis

Analysis of game dynamics provides additional insight into strategy behaviour. Table 4 summarises game length statistics across match ups.

Matchup	**Mean Turns**	**Median**	**IQR**
Naive vs Optimal	41.4 ± 17.0	38	[30, 49]
EV(λ) vs Naive	43.1 ± 15.5	40	[33, 49]
EV(λ) vs Optimal	36.5 ± 14.7	33	[27, 42]
DP-Median vs Naive	41.2 ± 16.9	38	[29, 48]
DP-Median vs EV(λ)	36.8 ± 15.3	33	[27, 42]
DP-Median vs Optimal	35.2 ± 15.1	31	[25, 41]

Table 4: Game length statistics (number of turns per game) across strategy matchups.

Games between sophisticated strategies (EV vs Optimal, DP-Median vs Optimal, DP-Median vs EV) are notably shorter, reflecting more efficient point accumulation when both players

employ near-optimal play. The turn outcome distribution reveals the underlying risk profiles of different matchups:

Matchup	Successful Holds	Busts	Catastrophes
Naive vs Optimal	53.4%	42.4%	4.2%
EV(λ) vs Naive	58.3%	37.9%	3.7%
EV(λ) vs Optimal	41.1%	53.5%	5.4%
DP-Median vs Naive	53.9%	41.9%	4.2%
DP-Median vs EV(λ)	41.3%	53.3%	5.4%
DP-Median vs Optimal	37.4%	56.9%	5.7%

Table 5: Turn outcome distribution across strategy matchups.

The higher bust rates in matchups against Optimal (53.5% for EV(λ) and 56.9% for DP-Median) reflect the aggressive nature of these strategies when competing against strong opposition. The DP-Median vs Optimal matchup shows the highest bust rate, indicating that even the simplified lookup-table approach must take substantial risks against the full dynamic programming solution.

3.1.2 Score Difference and Victory Margin Analysis

A deeper analysis of score differences reveals important patterns in how each strategy wins. Figure 11 shows the distribution of score differences when each strategy wins in the Naive vs Optimal matchup.

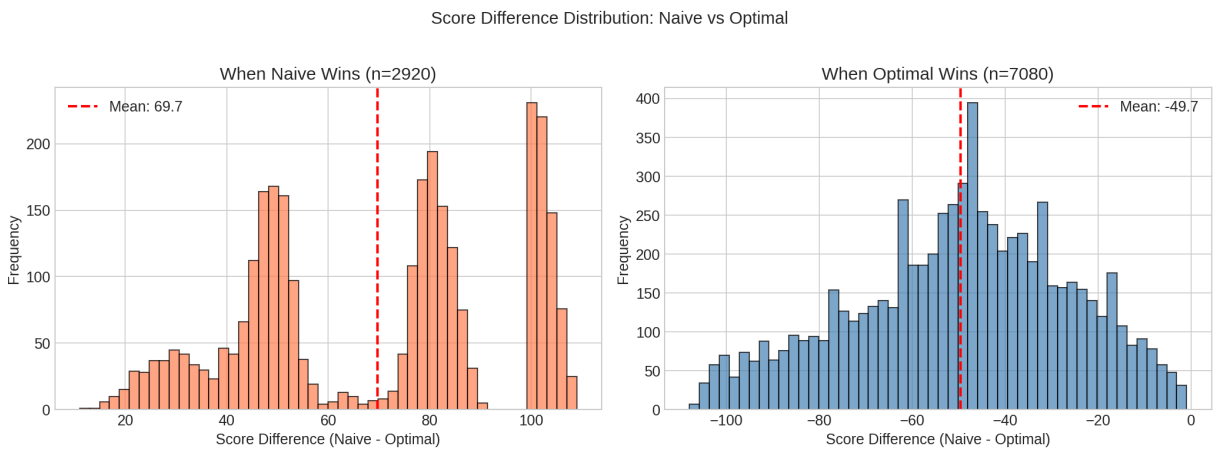

Figure 11: Score difference distribution in Naive vs Optimal matchup. Left: when Naive wins (trimodal distribution suggesting distinct failure modes for Optimal). Right: when Optimal wins (approximately normal distribution indicating systematic superiority).

A striking observation is the **trimodal distribution** when Naive wins, contrasted with the roughly normal distribution when Optimal wins. This pattern suggests that Naive can only defeat Optimal through specific failure modes:

- **Mode 1 (small margin)**: Late-game catastrophe for Optimal near victory

- **Mode 2 (medium margin)**: Mid-game catastrophe resetting Optimal's score
- **Mode 3 (large margin)**: Early or multiple catastrophes for Optimal

In contrast, Optimal's wins follow a normal distribution because it wins through systematic superiority: many small, consistent advantages that sum to a normal outcome via the Central Limit Theorem.

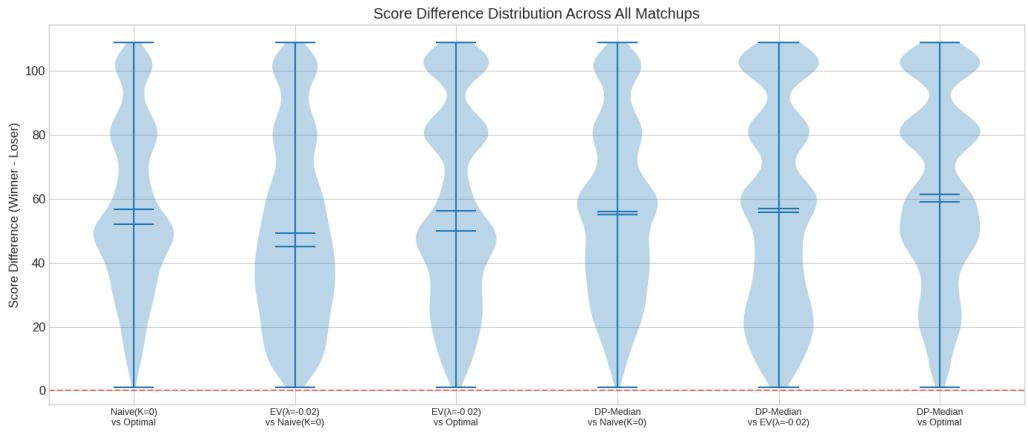

Figure 12: Violin plots comparing score difference distributions across matchups. The red dashed line indicates zero (tie). Wider regions indicate more common outcomes.

Figure 12 summarises the score difference distributions across all matchups. Naive vs Optimal shows a strong negative skew (Optimal dominates), while EV(λ) vs Naive and DP-Median vs Naive both show strong positive skews as these sophisticated strategies dominate the baseline. EV(λ) vs Optimal and DP-Median vs Optimal are relatively centred, reflecting balanced competition against the optimal strategy – we would expect this as the former strategy is derived from the latter. DP-Median vs EV(λ) shows a slight positive skew, consistent with DP-Median's 57% win rate in this matchup. Winners typically finish with scores between 100–110, while losers' final scores show high variance due to catastrophic double-six events.

3.2 Recommended Playable Strategy

Both Expectation and DP-Median offer simple rules that perform much better than the Naive baseline and remain close to the optimal DP strategy. In the head-to-head results, DP-Median shows slightly stronger and more stable performance, especially when playing against Expectation. Since its decision rules are straightforward and the full threshold table has already been presented in Section 5.1, DP-Median provides the best balance between practicality and playing strength. Based on these results, it is the strategy we recommend for actual game play.

4 Extension

4.1 Extension 1: More Than Two Players

This extension investigates the robustness of our derived strategies in a multi-player environment ($N > 2$). Due to the complexity introduced by the "double six" total-bust rule, a challenging benchmark was established where all non-primary opponents adopted the highly conservative **Hold at** $K = 0$ strategy, justified by prior analysis showing the optimal fixed-K threshold remains $\mathbf{K \leq 6}$ as competition increases.

We tested four advanced dynamic strategies for the primary player (Player 0) against this baseline, analysing win rates for $N = 3$ (simulated 50 times) and $N = 4$ (simulated 10 times):

- **Expectation Model:** Relies solely on Player 0's score ($\mathbf{t} = \frac{150-s}{11}$), demonstrating a score-independent strategy.
- **DP-Median Variants:** These adapt the optimal two-player strategy table by calculating the score difference (d) based on the opponents' score distribution:
  1. **Max:** d is calculated using the score of the maximum opponent.
  2. **Min:** d is calculated using the score of the minimum opponent.
  3. **Avg:** d is calculated using the average score of all opponents (The optimal variant).

The performance of the four tested strategies against the two $K = 0$ opponents is summarised in Table 6. The simulation confirmed the significant advantage of dynamic policies.

Strategy	**Win Rate (3-player)**	**Win Rate (4-player)**
Expectation	61.2%	55.4%
DP-Median (max)	63.8%	58.9%
DP-Median (min)	62.5%	57.2%
DP-Median (avg)	**64.4%**	**59.8%**

Table 6: Winning probability against $K = 0$ opponents in multi-player games. Random chance would yield 33.3% (3-player) or 25% (4-player).

The simulation showed the **DP-Median (avg)** strategy consistently yielded the highest performance, achieving a win rate of approximately **64.4%** in the 3-player game and **59.8%** in the 4-player game (far exceeding random chance). This result strongly suggests that basing the decision threshold on the **collective average threat level** is the most robust heuristic in multi-player settings. The strength of the (avg) variant confirms that the optimal choice involves proactively managing the heightened collective risk of all opponents. While the DP-Median (max) and (min) policies are only marginally lower, given the potential overhead of calculating opponent averages for very large player counts, these simpler max/min policies are highly recommended effective alternatives.

4.2 Extension 2: Single Die Roll

This extension uses a single die, which **entirely removes the "double six" (total score reset) rule**. The game becomes identical to the classic "Pig" game, with a single failure mode: rolling a 6 (Prob 1/6) busts the turn.

Without the total-bust risk, the optimal simple strategy is to maximise the expected score per turn. We can find this by balancing the expected gain of a "Safe Roll" (Prob 5/6, $E(\text{Safe}) = 3$) against the expected loss of a "Bust" (Prob 1/6, Loss $= t$).

$$E(\text{Gain}) > E(\text{Loss}) \quad \Rightarrow \quad \left(\frac{5}{6} \times 3\right) > \left(\frac{1}{6} \times t\right) \quad \Rightarrow \quad 15 > t$$

This calculation suggests a robust strategy is to **"Hold at 15"**. The full DP solution $T^*(S, R)$ would also be far less volatile than in the original "Sheep" game.

4.3 Extension 3: Fixed Turns and Highest Score Wins

This extension changes the objective from a "race to 100" to a **score maximisation** problem over a fixed number of turns (T). The core challenge of utilising the original game's rules is that standard, time-independent strategies perform sub-optimally when the game's horizon is known. Therefore, a strategy must adapt to maximise the final accumulated score after T turns.

Methodology: Dynamic Endgame Adjustment

While the standard **DP-Median** policy (H_{DP}), derived from the infinite horizon two-player game, provides a robust baseline, it is considered myopic in the crucial final turns. To address this, we developed a dynamic adjustment model to refine the Hold-at-Threshold (H) based on the remaining turns (k) and the current score difference ($d = s - o$). The refined decision rule is defined as:

$$H(s, d, k) = H_{DP}(s, d) + \Delta(k, d)$$

The baseline policy (H_{DP}) is the standard DP-Median strategy derived in Section 6.1. The Δ term applies a negative correction (more conservative) when leading and a positive correction when trailing.

Results and Strategic Recommendation

Through Monte Carlo search across short and medium horizons ($T \leq 15$ turns), we confirmed that introducing specific Δ adjustments based on the remaining turns significantly improves performance over the static baseline. The adjustments succeed by making the strategy more conservative when victory is near, and critically, more aggressive when trailing, which is essential for score maximisation.

Optimal Turn-Based Adjustments (Δ found via Simulation):

- **Early Game ($k \geq 5$):** Unmodified threshold ($\Delta = 0$).
- **Mid-Endgame ($3 \leq k \leq 4$):** Moderate correction ($\boldsymbol{\Delta}_{\mathbf{mid}} = \pm\mathbf{3}$).
- **Late-Endgame ($k \leq 2$):** Strong correction ($\boldsymbol{\Delta}_{\mathbf{late}} = \pm\mathbf{5}$).
- **Recommendation for Short Games ($T \leq 15$):** The dynamically adjusted DP-Median policy provides a superior strategic advantage.
- **Recommendation for Long Games ($T \geq 20$):** Our simulations revealed that for long fixed-turn games, these adjustments consistently reduce performance. Therefore, the **unmodified DP-Median rule is preferred** for long horizons.

5 Conclusion

This report analyses optimal decision-making in the sheep dice game, which is a two-player variant of the pig game, using two die and introducing a catastrophic risk, where rolling two sixes sets a players score to 0. We compare four strategies, increasing in complexity: the naïve fixed threshold strategy (K=0), expected value with a score difference term ($\lambda = -0.02$), dynamic programming optimal approach, and a playable take on the dynamic programming solution (DP-Median approximation).

Dynamic programming yields the exact optimal policy via value iteration over the state space (S, R, T), converging in 61 iterations. The first player wins 51.4 percent of the time. Monte Carlo simulations (n=10000) show that DP optimal defeats the naïve baseline with a 70.8 percent win rate, confirming its strategic advantage. Counterintuitively, we find that $EV(\lambda)$ achieves a 73.1 percent win rate when versing the naïve strategy, despite its lower game theoretic strength compared to the DP strategy, it takes fewer rolls per turn (1.74 vs 2.08), therefore suffering less catastrophic rolls, which exploits the naïve strategy's conservative play. Analysis of score difference distributions reveal that DP's losses follow a trimodal distribution, corresponding to 3 distinct failure scenarios, while for its wins we see a normal distribution, indicating DP's consistent strategic edge. The DP-median strategy approximates optimal play into a compact lookup table indexed by score phase and score difference. This strategy achieves near optimal performance, while remaining playable.

Extensions to multi-player games show that the DP-Median (avg) variant, which bases decisions on the average opponent score, achieves the highest performance (64.4 percent in 3-player games). Alternative game variants, including single-die rules and fixed-turn formats, require adjusted thresholds that reflect their modified risk structures. Future work should examine human adherence to the prescribed strategies and extend the framework to asymmetric objectives and alternative pig game variants.

6 Code and Data

The complete implementation code and simulation data for this project are available on Kaggle:

`https://www.kaggle.com/code/yushi520/math3060-project-1`
`https://www.kaggle.com/datasets/yushi520/dice-sims`

The repository includes the game simulation framework, strategy implementations, code used to visualise and run *most* simulations. Running the notebook with the provided data files will reproduce the statistics reported in this paper, up to a degree due to slight variation in aspects which require simulation, up until the extension.

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